

Technical Requirement Document

Here's a detailed Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-CUS-001

Related Functional Requirement(s): FR-CUS-001, FR-CUS-006

Objective: Design a RESTful API to create new customer records with email validation.

Target Cluster Configuration:

- Cluster Name: Customer Management Cluster
- Databricks Runtime Version: Not applicable (assuming a traditional web application)
- Node Type: Not applicable
- Driver Node: Not applicable
- Worker Nodes: Not applicable
- Autoscaling: Not applicable
- Auto Termination: Not applicable
- Libraries Installed: Email validation libraries (e.g., regex)

Source Data Details:

- Dataset Name: Customer Record Form Data
- Location: HTTP Request Body
- Format: JSON
- Description: Customer record details submitted by the user

Target Data Details:

- Output Name: Customer Record
- Location: Database (e.g., MySQL)
- Format: Relational data
- Description: Newly created customer record

Job Flow / Pipeline Stages:

- Validate email address format
- Check for duplicate records
- Create new customer record
- Log the "Create" operation

Data Transformations / Business Logic:

- Step: Email validation
- Description: Validate email address format using regex
- Transformation Logic: Use a regex pattern to match the email address format

Error Handling and Logging:

- Handle invalid email address format errors
- Log CRUD operations in a secure audit log

TR-CUS-002

Related Functional Requirement(s): FR-CUS-002

Objective: Implement a data retrieval mechanism to fetch existing customer records.

Target Cluster Configuration: Same as TR-CUS-001

Source Data Details:

- Dataset Name: Customer Records
- Location: Database (e.g., MySQL)
- Format: Relational data
- Description: Existing customer records

Target Data Details:

- Output Name: Customer Record Details
- Location: HTTP Response
- Format: JSON
- Description: Customer record details retrieved from the database

Job Flow / Pipeline Stages:

- Retrieve customer record details from the database
- Display customer record details to the user

Data Transformations / Business Logic:

- Step: Data retrieval
- Description: Retrieve customer record details from the database
- Transformation Logic: Use a SQL query to fetch the required data

Error Handling and Logging:

- Handle database query errors
- Log CRUD operations in a secure audit log

TR-CUS-003

Related Functional Requirement(s): FR-CUS-003, FR-CUS-006

Objective: Design an update mechanism for existing customer records with email validation.

Target Cluster Configuration: Same as TR-CUS-001

Source Data Details:

- Dataset Name: Updated Customer Record Form Data
- Location: HTTP Request Body
- Format: JSON
- Description: Updated customer record details submitted by the user

Target Data Details:

- Output Name: Updated Customer Record
- Location: Database (e.g., MySQL)
- Format: Relational data
- Description: Updated customer record

Job Flow / Pipeline Stages:

- Validate updated email address format (if changed)
- Update customer record
- Log the "Update" operation

Data Transformations / Business Logic:

- Step: Email validation
- Description: Validate updated email address format using regex
- Transformation Logic: Use a regex pattern to match the email address format

Error Handling and Logging:

- Handle invalid email address format errors
- Log CRUD operations in a secure audit log

TR-CUS-004

Related Functional Requirement(s): FR-CUS-004

Objective: Implement a deletion mechanism for customer records.

Target Cluster Configuration: Same as TR-CUS-001

Source Data Details:

- Dataset Name: Customer Record ID
- Location: HTTP Request
- Format: JSON

- Description: ID of the customer record to be deleted

Target Data Details:

- Output Name: Deletion Status
- Location: HTTP Response
- Format: JSON
- Description: Status of the deletion operation

Job Flow / Pipeline Stages:

- Prompt for confirmation
- Delete customer record upon confirmation
- Log the "Delete" operation

Data Transformations / Business Logic:

- Step: Deletion
- Description: Delete the customer record from the database
- Transformation Logic: Use a SQL query to delete the required data

Error Handling and Logging:

- Handle database query errors
- Log CRUD operations in a secure audit log

TR-CUS-005

Related Functional Requirement(s): FR-CUS-005

Objective: Implement a search mechanism for customer records by name or email.

Target Cluster Configuration: Same as TR-CUS-001**Source Data Details:**

- Dataset Name: Search Criteria
- Location: HTTP Request
- Format: JSON
- Description: Search criteria (name or email)

Target Data Details:

- Output Name: Search Results
- Location: HTTP Response
- Format: JSON
- Description: List of matching customer records

Job Flow / Pipeline Stages:

- Retrieve matching customer records from the database

- Display search results to the user

Data Transformations / Business Logic:

- Step: Data retrieval
- Description: Retrieve matching customer records from the database
- Transformation Logic: Use a SQL query to fetch the required data

Error Handling and Logging:

- Handle database query errors
- Log search operations (optional)

TR-CUS-006

Related Functional Requirement(s): FR-CUS-006

Objective: Implement email validation using regex.

Target Cluster Configuration: Same as TR-CUS-001

Source Data Details: Same as TR-CUS-001

Target Data Details: Same as TR-CUS-001

Job Flow / Pipeline Stages: Same as TR-CUS-001

Data Transformations / Business Logic: Same as TR-CUS-001

Error Handling and Logging: Same as TR-CUS-001

TR-CUS-007

Related Functional Requirement(s): FR-CUS-007

Objective: Implement logging for CRUD operations on customer records.

Target Cluster Configuration: Same as TR-CUS-001

Source Data Details:

- Dataset Name: CRUD Operation Details
- Location: Application logs
- Format: Log entries
- Description: Details of CRUD operations performed on customer records

Target Data Details:

- Output Name: Audit Log
- Location: Secure audit log storage
- Format: Log entries

- Description: Logged CRUD operations

Job Flow / Pipeline Stages:

- Capture CRUD operation details
- Store log entries in a secure audit log

Data Transformations / Business Logic:

- Step: Log capture
- Description: Capture CRUD operation details
- Transformation Logic: Use a logging framework to capture log entries

Error Handling and Logging:

- Handle logging errors (e.g., log storage unavailable)